Chapter 40

Derivatives Mishaps and What We Can Learn

This chapter is not a collection of cautionary tales. Every case in it is a natural experiment, one the profession never meant to run, and each one tested a specific assumption made somewhere earlier in this book.

One finding organizes the chapter and it is not the expected one. In almost none of these losses did a derivative behave in a way its model did not predict. The instruments performed. What failed was around them—the cash, the exit, the information, the venue, the buyer's ability to price what it bought, or the decision not to hedge at all. The last of those deserves saying at the start: the most recent case, Section 40.14, is a bank that failed because it did not hedge, and any chapter that presents derivatives as the source of danger has the sign wrong.

Chapters 38 and 39 built the measures and the rules; most of these losses fell in the gaps Section 38.19 lists, and Chapter 39's controls are read here against the cases behind them. Sections 40.1 to 40.14 take the cases one at a time; Section 40.15 draws out what repeats, Section 40.16 what changed, and Section 40.17 what institutions and regulators should take away.

40.1 METALLGESELLSCHAFT: CORRECT IN VALUE, WRONG IN TIME

In 1993 a German industrial group's US subsidiary had sold long-dated fixed-price oil supply contracts and hedged them with a stack of near-dated futures and short-dated swaps, rolled forward each month. The hedge was sound in the only sense a valuation model recognizes. Whether the parent was right to liquidate it rather than fund it is still argued, and Chapter 3's case study names both sides of that debate; what neither side disputes is that the cash the hedge demanded in a fall had not been committed in advance.

Example 40.1 The cash flow and the value point in opposite directions

Take the historical position, 160 million barrels of forward commitment, and treat it as hedged one-for-one with front-month futures; the real stack mixed exchange-traded futures with short-dated over-the-counter swaps. The price falls and the roll spread below are illustrative:

Price fall	Futures margin (cash out)	Supply contract (unrealized)
\$2.00/bbl	−\$0.32bn	+\$0.32bn
\$5.00/bbl	−\$0.80bn	+\$0.80bn
\$8.00/bbl	−\$1.28bn	+\$1.28bn

Every row is flat in undiscounted value and catastrophic in cash. The futures loss is paid in variation margin now; the supply contract's offsetting gain arrives over ten years and cannot be pledged. Chapter 3's case study records the historical outcome, margin calls of roughly \$0.9 billion by the end of 1993 and losses of about \$1.3 billion.

The roll was a second exposure no one had chosen. A stack-and-roll pays or earns the calendar spread every month on the whole remaining stack, which converts a flat-price hedge into a calendar spread position; Section 36.10 works that arithmetic. On this position, with deliveries running the stack down to an average of about 153 million barrels over the first year, a contango of $0.35 a barrel per roll costs about $0.64 billion that year while the flat-price hedge worked perfectly.

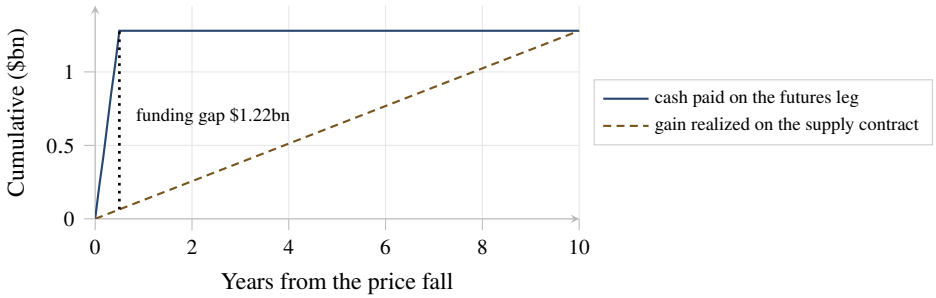


Figure 40.1: An $8/bbl fall spread over six months, on Example 40.1's position; illustrative.

Figure 40.1 is the mechanism of this chapter in one picture. Both curves reach $1.28bn, so the hedge is exact in undiscounted terms—and the vertical distance between them is a loan the strategy never arranged. The gap peaks at $1.22 billion at six months, once the margin has all been paid, and it closes only at the ten-year horizon.

Every case that Section 40.15 groups under cash and value is that gap. The LDI schemes, the 2022 European generators, the margin calls at AIG, and Example 37.2's basis trade all draw the same two curves with different labels. A valuation model computes where the curves end; the failure happens in between.

The lesson is Section 8.9's: margin calls are a liquidity risk, and Section 8.10's stress test measures them because it counts gross cash outflow, not net loss. On Example 40.1's position it reports $0.16 billion of variation margin for every dollar the oil price falls, where a value measure reports nothing. The funding line should be committed at that scale before the trade, and the roll budgeted as a calendar spread.

40.2 PORTFOLIO INSURANCE, 1987: WHEN THE HEDGE IS THE FLOW

This is the clearest case in the chapter of a hedging technique moving the market it was hedging against (the LDI selling of Section 40.8 is another). Section 16.2 introduced the technique; this is the episode and its arithmetic.

Portfolio insurance replicated a protective put synthetically: rather than buy a put, sell index futures as the market falls and buy them back as it rises, in the proportion Chapter 18's delta prescribes (Sections 18.11–18.12). The replication is sound in theory, and the programs followed it as closely as the market allowed (on October 19, 1987 they fell behind schedule because the futures market could not absorb the selling), so the mechanism failed, not the execution.

Example 40.2 The selling schedule was published by the mathematics

Take $70 billion insured with a floor 5% below spot over three months at 20% volatility, with the interest rate set to zero for simplicity. The programs must be short $-\Delta$ of the index on a portfolio whose own value falls with the index, so the futures short is $|\Delta| \times \$70\text{bn} \times S/100$ at index level S:

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Index	Put delta	Futures short	Futures sold
100	−0.287	$20.07bn	—
95	−0.480	$31.92bn	+$13.18bn
90	−0.688	$43.35bn	+$13.48bn
80	−0.952	$53.33bn	+$15.89bn

A 10% decline obliges the programs to sell $26.7 billion, and a 20% decline $42.5 billion, with no discretion at any point. That is more than the change in the short, because the contracts already held lose dollar value with the index, and that loss needs no trade.

The selling per point of the index is driven by gamma, so it is fastest near the floor and slows once the delta approaches −1; the last row covers ten points and the two above it five each. Near the floor the delta changes by about 0.042 per index point, which, on the 0.70 billion index units insured at an index of 95, is about $2.79 billion of futures per point. Figure 40.2 draws the whole schedule.

The critical feature is what a synthetic put changes. A purchased put has a *seller*. Left unhedged, that seller's position calls for no trading at all; a dealer who delta-hedges the short put is short gamma and *sells* into the decline as well, as Section 18.12 shows. Listed options therefore spread the selling across writers, not all of whom hedge; they do not abolish it. Replication puts the whole of it in the holders' own programs, on a schedule anyone can compute in advance. The protection was real for each individual holder and, in aggregate, was manufactured out of the very selling that deepened the decline it insured against.

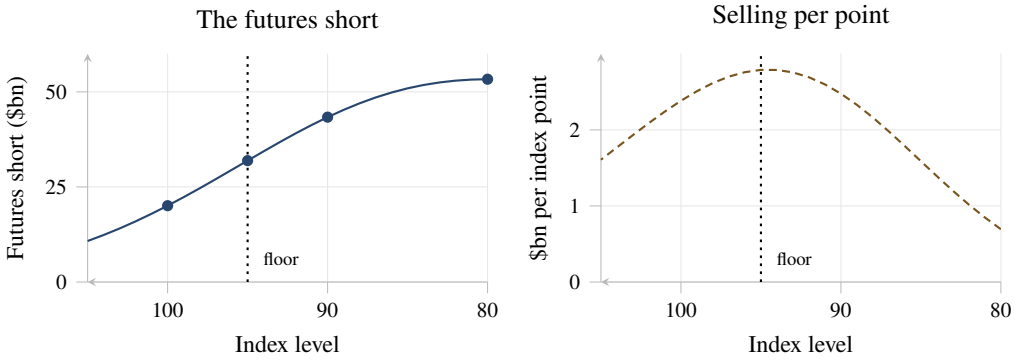


Figure 40.2: The portfolio insurers' selling schedule in Example 40.2, as the index falls from 105 to 80.

The mechanism has appeared three more times in this book. Section 35.6's autocallable issuers buy into a decline and then, at the barrier, sell about a fifth of the hedge at once; Section 37.3's liquidation cascade forces sales as a published function of price; and Section 23.7's dealer gamma positioning is the same arithmetic run daily on a much shorter horizon. In each, hedging demand is a mechanical function of the price, so at the level that matters the hedge and the move reinforce each other. That is the procyclicality of Section 38.18 with a hedge ratio in place of a risk measure: there a capital requirement that rises after volatility arrives forces positions to be cut into a falling market, and here a delta that grows as the index falls does the same. In both, the feedback needs the rule to be shared. One program selling on its own delta moves nothing; $70 billion of insured portfolios following the same rule at once moved the index, and the move set the next round of selling.

What changed afterward is worth noting because it was not a ban. Circuit breakers were introduced (a mechanism that interrupts the feedback rather than the strategy), and the listed index option market grew. The fix was to make the protection purchasable rather than

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manufacturable, and that is the correct response to a mechanism rather than to an instrument.

Section 18.12's gamma arithmetic publishes a replicating hedge's flow in advance: the futures sold per index point (gamma times the insured portfolio's value) peaks in Example 40.2 at about $2.79 billion at the floor (Figure 40.2). A program should set that number, summed over everyone running the same rule, against what the futures market can absorb, and where the market cannot absorb it, buy a listed put instead.

40.3 BARINGS AND SOCIÉTÉ GÉNÉRALE: A FAILURE OF CONTROLS, NOT OF DERIVATIVES

Two cases, thirteen years apart, are failures of the risk control system rather than of the instruments, and neither has any derivatives pricing content. In both, the checks that should have caught an unauthorized position (the **segregation of duties** that keeps trading apart from its settlement, independent reconciliation, and verification of what the trader reported) were controlled or defeated by the person they were meant to check.

In both, positions reported as low-risk arbitrage were outright directional. At Barings one person controlled trading and the settlement of that trading, and performed the reconciliation that would have exposed the position. At Société Générale the trader had come from the middle office and knew its checks well enough to hide the positions behind fictitious offsetting trades.

	Position	Firm equity	Ratio
Barings, 1995	$27bn	$0.6bn	**45**×
Société Générale, 2008	EUR 50bn	EUR 30bn	1.7×

The same failure destroyed one firm and did not destroy the other, and what decided the outcome was the ratio. Barings held about forty-five times its equity in directional futures and option positions and lost £827 million on them; the French bank held a larger notional against a balance sheet that could absorb the EUR 4.9 billion it cost.

The instruments were exchange-traded: index futures in both, and at Barings also Nikkei options and Japanese government bond and Euroyen futures, all cleared, margined daily, and priced as in Chapters 5 and 15. No model failed and no instrument surprised anyone.

Separating trading, settlement, and reconciliation is necessary and will still sometimes fail, and when it does, the severity is set by a number the controls do not watch: the largest possible position against equity. Section 2.3's daily settlement lets a firm cap that number, because every exchange-traded loss is paid in margin the next morning, and capping the margin funding a desk can draw caps the position. At Barings that funding was the visible signal: the head office kept remitting margin to Singapore for a position it believed was hedged.

40.4 LTCM: LEVERAGE PRICES THE PATH, NOT THE VIEW

The 1998 failure of Long-Term Capital Management (LTCM) is usually told as a story about overconfidence in models: a fund whose partners included two Nobel laureates trusted spreads and correlations estimated in calm markets, and the Russian default of August 1998 broke them. That story is not wrong, but the arithmetic tells a narrower and more useful one.

Example 40.3 A correct view at the wrong leverage

Equity of $4.7bn against a $125bn balance sheet is **26.6**× leverage, before derivatives notionals far larger than either.

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Adverse move on assets	Loss	Share of equity
0.5%	$0.63bn	13%
1.0%	$1.25bn	27%
2.0%	$2.50bn	53%
4.0%	$5.00bn	**106%**

The wipeout move is $4.7bn/$125bn = 3.76%. So the strategy could fail with the convergence trades (long the cheaper of two closely related securities, short the richer, waiting for the gap to close) right: it needed them to converge without the balance sheet first losing 3.8% of its value. And convergence spreads often widen before they converge; bearing that widening is what the trade is paid for. At that leverage the fund was not expressing a view about value; it was expressing a view about the path, that spreads would not widen that far before converging, and it had not priced that path.

Leverage converts the path into a survival condition that a VaR at a horizon does not see, so size the trade by the path: Section 38.17's reverse stress test finds the adverse move the position must survive on the way to converging, 3.76% of the balance sheet in Example 40.3, and leverage should keep that move well beyond the widening such trades have shown. A position others also hold makes the move larger: the fund's forced sales mark down the others' books and they sell too, which is Section 40.2's shared rule in another form. LTCM's trades had been copied across the dealer community and were unwound together in the summer of 1998; on September 23 the New York Fed brokered a $3.6 billion recapitalization by fourteen firms.

40.5 AMARANTH: EXIT COST IS A FUNCTION OF YOUR OWN SIZE

The hedge fund Amaranth's natural gas loss of about $6.6 billion in September 2006 shows that concentration, which Section 38.19 says a distribution-based measure cannot see, is not an academic gap. The fund held positions approaching 40% of open interest in NYMEX's winter-month natural gas contracts. At that share, the exit is not a market price—it is a schedule. With daily volume taken, for illustration, as equal to the open interest of those contracts, the days to exit are 40% divided by the participation rate:

Maximum participation in daily volume	Days to exit
10%	4.0 days
20%	2.0 days
30%	1.3 days

Every one of those days is a day the market knows a forced seller is present. VaR is read from the distribution of price changes, which does not depend on how much of the market the holder owns; only the exit does.

Set the horizon of the measure by the holder's own exit, not by the risk factor's class. At a 10% participation rate the table's exit takes four days, so Section 38.14's square-root-of-time rule scales the one-day VaR to the four-day exit by a factor of two if the position is held whole until sold—and it still assumes prices unmoved by the seller, which at 40% of open interest they are not. A risk manager should therefore limit the share of open interest directly and stress the exit with an impact cost, since Section 39.4's liquidity horizons, assigned by class, will not.

40.6 ARCHEGOS: THE SAME RISK AT MANY BROKERS

In March 2021 Archegos Capital Management, a family office holding concentrated stock positions through swaps, collapsed: its capital was wiped out, and the dealers lost more than

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$10 billion. It was a failure of measurement, not of modeling.

Example 40.4 Adequate margin against the wrong exposure

Take a family office holding concentrated single-stock exposure through total return swaps at six prime brokers. Each broker saw its own slice and margined it at a level it judged adequate for that slice. Section 27.14 works the per-broker arithmetic, which leaves $7.5 billion of margin across the system against a $50 billion gross position.

But the liquidation was not six independent liquidations. All six sold the same stocks within days, and since a broker could look only to the margin it held, only the margin stood against the loss:

Impact of simultaneous exit	Loss	Against $7.5bn margin
10%	$5.0bn	covered
25%	$12.5bn	**short by $5.0bn**
40%	$20.0bn	**short by $12.5bn**

The margin was set for the exposure each broker measured and was wrong for the exposure that existed. No broker's own book showed the difference, and Section 27.14 adds that the margin levels themselves were set by competition for the client's business rather than by the concentration of the positions.

The instrument choice was not incidental. A total return swap gives economic exposure without an ownership disclosure, so the aggregate was invisible by construction: the same property that made the product attractive made the risk unmeasurable. Section 27.13 argued that bilateral measurement has a limit, and this is the limit.

Margin a total return swap (Section 25.9) on the client's whole position and its exit, not on one broker's slice: Section 27.15's initial margin is meant to cover the close-out loss, and in Example 40.4 a joint exit at 25% impact loses $12.5 billion against $7.5 billion collected. A prime broker should ask for the positions held elsewhere, treat a refusal to disclose them as a risk in itself, and set the rate by Section 40.5's exit arithmetic, not by competition.

40.7 LME NICKEL: WHEN THE VENUE IS THE RISK

In March 2022 the nickel price, which had closed near $29,000 a metric ton on Friday, March 4, more than tripled in the next two trading days to above $100,000, against a very large short position held by a producer as a hedge. Take an illustrative 100,000-metric-ton short; the producer's actual position was larger, and much of it was held over the counter, out of the exchange's view.

Price	Loss on a 100,000-metric-ton short from $29,000
$40,000	$1.10bn
$50,000	$2.10bn
$80,000	$5.10bn
$100,000	$7.10bn

The exchange then canceled several hours of trades. That cancellation is the case's real content, and it tests the assumption Section 2.10 flagged as a judgment about institutional resilience rather than a fact. Every calculation in Parts I and V treats the clearing house as the counterparty that does not fail, which is why central clearing improves on bilateral trading. Here the exchange and its CCP faced the prospect of member defaults they judged could exceed the resources standing behind the clearing house, and the exchange's resolution was to void executed trades. No member defaulted, and the default waterfall was never drawn on.

So the risk that materialized was not the price. A trader who was long nickel, correct, and fully margined had the profit taken away, which the reader met in a different market as

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Section 37.10's auto-deleveraging, and which Chapter 2's account of this episode records as trade finality yielding to the clearing house's survival.

A CCP's guarantee is bounded by its resources and its governance, and when they are threatened it can become a negotiation; Section 2.8's waterfall has a bottom, and this case showed that an exchange may act before its clearing house gets there. A member or end user should measure each layer of that waterfall against a move like March 2022's and read the venue's rules on canceling trades with it, which is Section 2.10's question of who absorbs the loss; the venue should know the full size of a large position, over-the-counter holdings included.

40.8 UK LDI: THE HEDGE THAT CONSUMED ITS OWN COLLATERAL

In September 2022 a rapid rise in gilt yields forced UK pension schemes into a self-reinforcing sale of the assets backing their hedges. What makes the case valuable is that the schemes' hedges did their job.

Example 40.5 Better off and unable to pay

Take a system with roughly £1.5tn of liabilities at a duration near twenty, hedged with leveraged gilt and swap positions requiring variation margin. The schemes' reported hedge ratios were higher than the 40% used here, but part of each hedge was gilts held outright, which call for no collateral. Take the margined part as covering 40% of the liabilities' rate sensitivity, so each margin call is 40% of the fall in the liabilities' value:

Yield rise	Liability falls by (duration held at 20)	Margin call
50 bp	10%	£60bn
100 bp	20%	£120bn
150 bp	30%	£180bn
250 bp	50%	£300bn

The funding position improved at every row. Because the hedge covered only part of the liabilities' rate sensitivity, the liabilities fell faster than the assets, so the schemes became *better funded* as the crisis progressed—while unable to meet the margin calls.

Selling gilts to raise the collateral pushed yields higher, which raised the next margin call. The hedge's own success generated the demand that forced its unwinding.

This is Section 8.9 in its purest form—solvency and liquidity are different quantities, and only one of them forces a sale—and the lesson is Section 40.1's. The margined hedge in Example 40.5 is a DV01 of $0.40 \times £1.5\text{tn} \times 20 \times 0.0001 = £1.2$ billion per basis point, so Example 8.10's arithmetic makes a 100 basis point move a £120 billion margin call, and Section 38.17's reverse stress test, run on the buffer a scheme holds, finds the move that exhausts it—the figure to report. Plan for a larger move than any one scheme's stress, because forced gilt sales raise yields again: Example 8.11's loop, whose total only a supervisor can see.

40.9 AIG: A COLLATERAL TRIGGER ON A POSITION THAT NEVER DEFAULTED

By the end of 2007, according to AIG's own annual report, its financial products unit had sold protection on about $527 billion of super-senior tranches, on a view that they attached above any plausible loss. Most of it was written for European banks seeking regulatory capital relief; the margin calls came mainly from about $78 billion written on multi-sector CDOs backed largely by subprime mortgages. The view was defensible on realized defaults for much of the book, and it was irrelevant, because the contracts contained two features the view did not price:

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mark-to-market collateral against spread widening, and additional collateral on a downgrade of the seller.

Spread widening	Collateral demanded on $78bn (duration held at 5)
100 bp	$3.9bn
200 bp	$7.8bn
400 bp	$15.6bn
800 bp	$31.2bn

No row requires a default. The position was a view on realized losses; the contract was a claim on marks, and Section 25.4's mark moves with the spread, default or not.

The downgrade trigger is unlike anything in an ordinary hedge, whose cash demands depend on the market and not on the hedger's own credit. It converts a deterioration in the *seller's own* condition into an immediate cash demand, so the contract becomes most expensive exactly when the seller is least able to pay. That is Section 27.9's wrong-way risk written into the contract rather than arising from a correlation. In September 2008 the downgrade came, the calls could not be met, and the Federal Reserve lent AIG $85 billion.

Price the collateral terms, not only the view on defaults. Section 25.4's mark makes the collateral a duration calculation—$78 billion at a duration of 5 posts $39 million for every basis point of widening, with no default—so the seller's liquidity should be sized by Section 8.10's stress test applied to spreads, not by expected loss. The downgrade trigger belongs in the same test, assumed to arrive with the widening that produces it.

40.10 THE LONDON WHALE: THE MEASURE CHANGED AND THE RISK DID NOT

In 2012 a credit index book inside a bank's treasury function lost about $6.2 billion. The instructive part happened months earlier and cost nothing at the time.

Reported VaR under the previous model	$132m
Reported VaR under the new model	$66m
Reduction	**50%**
Positions changed on the day	**none**

A model change, hastily approved and containing a spreadsheet error, halved the desk's reported risk without anyone trading. Over the following months the book's notional roughly tripled under the looser limit. The eventual loss was 47 times the old figure and 94 times the new one—a one-day figure set against a cumulative loss, but the scale is the point.

This is Chapter 39's model risk in its purest administrative form. Section 39.14 argued that an error in the model misstates the mark and the measure together; here the measure moved alone, and the limit framework built on it loosened accordingly. A limit expressed in units of a model is a limit the model can relax.

This case also most directly justifies Section 39.6's attribution test. The desk's risk model stopped describing its book, and the mechanism that now exists to detect that (comparing the P&L the risk model produces with the desk's own revaluation, day by day) is exactly the test this case calls for. Whether it would have caught this one is unclear, since Example 39.5 shows a desk can leave about a third of its P&L variance unexplained and still pass.

Validate a VaR model change as Section 39.14 validates any model, before the new number reaches a limit: run old and new side by side, require the fall from $132 million to $66 million on an unchanged book to be explained, and ask for Section 38.16's backtest on both. A limit the model cannot relax, on gross notional, should sit beside the VaR limit, so that a book tripling in size is visible whatever the measure says.

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40.11 COMPLEXITY SOLD TO A BUYER WHO COULD NOT PRICE IT

A fifth mechanism runs through several cases and is different in kind from the four that Section 40.15 draws from the cases so far, because the loss is not caused by cash, size, information, or a venue. It is caused by an asymmetry in who can value the trade.

Example 40.6 Two leverages multiplied

Orange County, California, ran a $7.5 billion municipal investment pool, levered it to about $20.5 billion through repo (**2.73**×) and invested a large share of it in **inverse floaters**, whose coupon falls as rates rise.

The instrument carries its own leverage. An inverse floater with coupon leverage L, paying a fixed rate less L times the floating rate, has roughly $(1 + L)$ times the duration of its underlying, taken here as 4 years, so the two multiply. Were the whole book in such notes:

Coupon leverage	Instrument duration	Portfolio effective duration
1	8.0 yr	21.9 yr
2	12.0 yr	**32.8 yr**
3	16.0 yr	43.7 yr

The actual book mixed inverse floaters with other structured notes and plain agency paper, so its duration was lower than any row above. Its 1994 loss of about $1.64 billion, on a rise of roughly 300 basis points in short and intermediate yields, corresponds to an asset duration of $1.64/(0.03 \times 20.5) \approx 2.67$ years, an effective duration of $1.64/(0.03 \times 7.5) \approx 7.3$ years on the depositors' $7.5 billion:

Rate rise	Loss	Share of the $7.5bn
100 bp	$0.55bn	7.3%
200 bp	$1.09bn	14.6%
300 bp	$1.64bn	**21.9%**

Nothing was concealed. The coupon formula was in the term sheet and the repo was on the balance sheet. The pool reported its yield advantage rather than its duration, warnings during 1994 that further rate rises threatened it were dismissed, and the product was described, accurately as far as its yield went, as an income enhancement. The county filed for bankruptcy in December 1994.

The same asymmetry produced the structured-swap disputes of the mid-1990s, in which corporate treasurers bought leveraged rate positions embedded in what were presented as financing trades, and later argued in court that they had not understood what they held. The legal outcomes varied; the mechanism did not.

It is not a historical category. Chapter 35's retail structured products carry the same asymmetry in a regulated wrapper, Section 35.10's estimated initial value exists precisely to quantify it, and Section 37.8's liquidity providers are selling a straddle described to them as yield. The mechanism persists because the asymmetry is structural: the party that constructs a payoff is usually better placed to price it than the party that buys it.

The response to this category is disclosure of the computed risk, not only its inputs, and suitability, because no risk system helps a buyer who does not know which risk to measure. The measure Orange County needed was Section 4.11's duration taken through the whole structure: about 2.67 years on the assets times repo leverage of 2.73 is Example 40.6's 7.3 years on the depositors' money, so each 100 basis points of rise costs 7.3% of the pool. A buyer should not hold a payoff whose duration it cannot compute, and Sections 35.11 and 35.12 set out how a note is evaluated and what disclosure cannot fix.

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40.12 EUROPEAN ENERGY 2022: THE SAME QUARTER, A DIFFERENT INDUSTRY

The LDI episode was not the only funding crisis of that quarter, and the second one matters because it suggests the mechanism is structural rather than a feature of pension funds. A generator that has sold power forward holds a short position that is margined daily. The plant that will deliver the power is not. Example 8.9 works the arithmetic for one producer; at the scale of a large generator, 50 TWh of forward sales and a move from 50 to 400 EUR/MWh demand EUR 17.5 billion of cash against a position whose entire purpose was to remove risk. The generator's value rose throughout (its plant's unsold output now sells into a far higher market), yet several such firms required state liquidity support to avoid failing. The support went to the firms that had hedged: those that had not faced no margin calls and took windfall profits as prices rose, which is what a system produces when it prices risk correctly and liquidity not at all.

This case bears on a question the earlier cases leave open. Metallgesellschaft could be dismissed as a strategy error and the LDI schemes as a leverage error. Two simultaneous instances in unrelated markets suggest the exposure is created by the market structure: it arises wherever a hedge is margined and the hedged item is not. Neither 2022 case was a product of mandatory clearing: most of the utilities hedged on exchanges by choice, and the pension schemes, exempt from the clearing mandate, met margin calls on repo and bilaterally margined swaps. But the post-2008 push toward margining, voluntary and mandatory, that Chapters 2 and 8 describe widened the footprint of that shape.

The lesson is Section 40.1's again, at the level of an industry. A forward sale of 50 TWh posts EUR 50 million of variation margin for every euro the power price rises, so the liquidity a generator commits before hedging should be sized by Section 8.10's stress test on a move like 2022's, not a typical month. A policymaker who pushes hedging into margined form should ask where that liquidity comes from in a stress.

40.13 MARK-TO-MODEL, AND POSITIONS WITH NO MARKET

A further mechanism applies to positions with no market price. Section 39.11's prudent valuation exists because the value carried for them is produced by the holder; this section adds the cases.

The failure mode is not fraud, though fraud is available. It is that a valuation that cannot be checked is a valuation that responds to pressure, and the pressure is asymmetric: no one objects to a mark that produces a profit.

The structural cases are two: Enron's long-dated energy contracts, booked at their full modeled value on the day of signing, which converted a decade of anticipated profit into current-period earnings; and illiquid structured credit carried at model prices while the market for it disappeared. In both, the position and the price came from the same place.

The safeguards are Chapter 39's. Independent price verification exists because the desk cannot be the source of its own marks. The model risk component of Example 39.9's adjustment, the largest single item, is a deduction for precisely this, and it is taken whether or not anyone suspects a problem.

The connection to machine learning is direct. A neural-network pricer's silent extrapolation outside the range of its training data (the training box of [TN41]) is mark-to-model with the model's confidence removed from view, which is why Section 39.15 insists the training set be recorded as part of the model.

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40.14 SILICON VALLEY BANK: THE LOSS FROM NOT HEDGING

This is the most important case in the chapter. A bank funded by concentrated, uninsured, rate-sensitive deposits held a large securities portfolio at meaningful duration and left the interest rate risk unhedged, having unwound during 2022 most of the pay-fixed swaps it had held against part of the book.

Example 40.7 What a swap would have cost, and what its absence did

Take a $120bn portfolio at duration 5.6:

Yield rise	Economic loss (duration held at 5.6)	Share of a $16bn equity base
100 bp	$6.7bn	42%
200 bp	$13.4bn	84%
300 bp	$20.2bn	**126%**
400 bp	$26.9bn	**168%**

A pay-fixed swap on about half the duration would have cost negative carry (in 2021, paying a fixed rate near 1% against an overnight rate near zero, on the order of 100 basis points a year on the swapped notional) and would have removed about half of that column. The instrument was standard, liquid, cleared, and priced by Chapter 7's arithmetic. Nothing about the hedge was difficult.

Accounting completed the failure. Securities classified as held-to-maturity are not marked through capital, so the reported number stayed intact while the economic number did not. The unrealized loss, about $15 billion at the end of 2022, was visible in the footnotes but not in capital, until, in March 2023, a loss-making sale of almost all its available-for-sale securities and a capital raise put it in front of depositors at once, by showing that the bank needed cash.

So the sequence is: an unhedged exposure, an accounting treatment that concealed it, and a funding base that could leave in a day. Each is necessary; none alone is sufficient.

A bank failed because it did not hedge its rate risk, and the derivative it did without was the plainest instrument in this book. Section 3.2's case for hedging applies in full, and it is worth teaching against the reflex that derivatives create danger.

Banking-book rate risk sits outside Chapter 39's trading-book rule. The Basel Committee's 2016 standard treats it in Pillar 2, with a supervisory outlier test: a change in the economic value of equity, under prescribed rate shocks, above 15% of Tier 1 capital. It looks through the reported number to the economics, but the US agencies never adopted it as a rule, and it did not bind SVB.

Hedge the economic duration with the tools of Sections 4.11 and 6.7: $120 billion at a duration of 5.6 is a DV01 of $\$120\text{bn} \times 5.6 \times 0.0001 \approx \$67$ million against $16 billion of equity, and a pay-fixed swap valued by Section 7.3's arithmetic and sized to half of it would have halved every row of Example 40.7's table. Measure that DV01 with held-to-maturity securities included, whatever the accounting says: as Section 3.8 shows, the swap would have been marked through earnings while the held-to-maturity securities were not, so the reported numbers would have penalized the hedge.

40.15 WHAT REPEATS

Across the cases of this chapter, four mechanisms recur and the instruments do not.

Cash and value are different quantities. Metallgesellschaft, the LDI schemes, the 2022 European generators, AIG, and Example 37.2's basis trade are one mechanism: a position correct in value can be closed for want of cash, and no valuation model contains the closing.

Size and leverage change the exit. LTCM, Amaranth, Archegos, and the nickel case all failed at the point of liquidation rather than at the point of valuation. The measure describes the

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position's distribution; the loss came from the position's share of the market, or from leverage that made the path decisive, which are different variables.

Someone must see the whole. Barings, Société Générale, and Archegos are cases where the aggregate existed and no one charged with controlling it saw it (inside a firm in the first two, across firms in the third). The failure is informational, and in Archegos the instrument enabled it.

And the guarantee has a bottom. The nickel case and Section 37.10's auto-deleveraging both show that the mechanism meant to make a market safe is bounded: at the exchange, trades were voided before any resource was drawn on, and at a crypto venue, profitable positions are closed once the insurance fund is gone.

A fifth mechanism sits apart. Section 40.11's asymmetry is not a failure of measurement but of *who is equipped to measure*, and it is the only one here that recurs in regulated retail form. Section 40.13 adds a sixth: a mark that only the holder can produce.

Case Study 40.1 The information was already inside the firm

The most consistent finding across the official inquiries is also the least comfortable, because it rules out the easiest explanation, that no one could have known.

In almost every case the facts needed to prevent the loss existed inside the institution before it happened. Barings' reconciliation discrepancies were reported and unresolved for months. The exchange-traded part of the nickel short was visible to the exchange, though much of the position was held over the counter. The credit index desk's own valuations were disputed internally before the loss was recognized. The unhedged securities portfolio of Section 40.14 had been the subject of supervisory findings. The Archegos post-mortem commissioned by Credit Suisse is explicit: the bank's risk function had escalated the concentration, and the escalation did not change the position.

So the failure is rarely detection and usually escalation. Someone saw it; the observation did not reach a person with both the authority and the incentive to act, or reached them in a form that competed with a revenue argument and lost.

That is why so little of the reform in Section 40.16 is quantitative. Segregating trading from settlement, requiring mandatory leave, making risk report outside the business line, and giving a desk-level test a capital consequence are all attempts to change what happens to an observation, not to generate a better one.

The corollary for a reader of this book is uncomfortable. Improving a measure has no effect if the existing one was already overridden, and much of Chapter 39's machinery, the backtesting zones and the attribution test with their capital consequences, tries to make overriding expensive rather than measurement better. That is the right design.

40.16 WHAT ACTUALLY CHANGED

The reforms these cases produced follow an unflattering pattern. The control failures produced durable change. Segregation of trading from settlement, independent reconciliation, and mandatory leave became standard after Barings, and such losses are rarer, though Société Générale in 2008 and UBS in 2011 (a $2.3 billion unauthorized-trading loss) showed that controls are not self-enforcing. This, the profession's clearest success, required no modeling.

The clearing reforms after 2008 were substantial and are examined in Chapters 2 and 27. They belong to a wider shift, though, which also widened the exposure of Sections 40.8 and 40.12: margining more of the market, voluntarily as well as by mandate, converts credit risk into liquidity risk, and the 2022 LDI and energy episodes showed the cost of that conversion.

The measurement failures produced the least. Amaranth's lesson about concentration was

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available in 2006, and Archegos repeated it in 2021 with a different instrument. Chapter 39's framework prices illiquidity and still does not measure concentration.

The accounting problem behind Section 40.14 is unresolved. The treatment that concealed the loss remains available, and the arguments for it are not frivolous: marking a held-to-maturity book through capital would import volatility into the accounts of institutions that genuinely intend to hold. The gap is real and the fix is not obvious, so the case will recur.

40.17 WHAT INSTITUTIONS AND REGULATORS SHOULD TAKE AWAY

Above the cases sits one rule: separate the instrument from the failure. In every case but nickel, where the venue voided the contract, the derivative did what its contract said, and a risk function that blames the instrument will look for the next problem in the wrong place. The failures were in cash, size, information, valuation, and governance, and the advice follows them.

Six points apply to a financial institution:

- *Measure liquidity beside value.* Report, for every hedged position, the margin it can call under stress against the liquidity committed to meet it, gross and never netted against the hedged item (Metallgesellschaft, LDI, AIG, the 2022 generators).
- *Size positions by their exit.* Limit the share of open interest and the client's whole position across brokers, not only the notional or the VaR, and stress the exit with a price-impact cost (Amaranth, Archegos, nickel).
- *Cap what a control failure can cost.* Keep the segregation of duties, and cap the margin funding any one desk can draw, so that the largest position that could be hidden is small against equity (Barings, Société Générale).
- *Separate who marks from who profits.* Verify prices independently, reserve for model risk, and validate a risk-model change before it sets a limit (the London Whale, Section 40.13).
- *Start from the failure.* Run a reverse stress test for the move that exhausts equity or liquidity, and assume others hold the same position (LTCM, portfolio insurance, LDI).
- *Treat removing a hedge as a decision.* Leaving a risk unhedged is a position, and taking a hedge off deserves the review that putting it on received (SVB).

Six apply to a regulator:

- *Look at the aggregate.* Margin can be adequate in every bilateral relationship and inadequate in total; position reporting across prime brokers and supervisory review of concentration cover what no single firm sees (Archegos, Amaranth).
- *Supervise liquidity as closely as capital.* Margining converts credit risk into liquidity risk, so buffers should be sized to stressed margin calls, including at non-bank users such as pension schemes and utilities (LDI, the 2022 generators).
- *Interrupt feedback rather than ban strategies.* Where many holders follow the same rule, circuit breakers, margin that does not fall in calm markets, and backstops prepared in advance cut the loop (portfolio insurance, LDI).
- *Make a venue's guarantee, and its limits, known.* Test default resources against plausible extreme moves, and publish the rules for canceling trades before they are needed (nickel).
- *Protect the buyer who cannot price.* Require disclosure of duration, leverage, and a payoff scenario table, and a suitability test on the payoff's shape (Orange County).
- *Look through the accounting.* Supervisory measures should capture economic duration and unrealized losses, whatever the accounting classification (SVB).

Five apply to governance and culture, at institutions and supervisors alike:

- *Question an unexplained profit as hard as an unexplained loss.* A book reported as

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low-risk arbitrage that earns like a directional one is directional, and no one objects to a mark that produces a profit (Barings, Société Générale, Section 40.13).

- *Fix escalation, not only detection.* Give the risk function an independent line to the chief executive and the board, and do not let the desk settle a risk manager's objection, because the concern had usually been raised and lost (Archegos, the London Whale).
- *Pay as the risk is realized.* Defer and claw back pay, and judge a desk on profit net of the risk and capital it uses, so that neither a decade of modeled profit booked at signing nor a halved VaR is rewarded (Enron, the London Whale).
- *Approve a position again as it grows.* A book approved at one size is a different book at three times that size, so growth past a set threshold should send it back to whoever approved it (the London Whale, Archegos).
- *Name an owner for each risk, and protect whoever raises it.* A discrepancy reported and left unresolved for months needs a named senior manager answerable for closing it, and the person who reported it needs protection from the revenue argument it contradicts (Barings, the case study in Section 40.15).

SUMMARY

In almost none of these losses did a derivative behave in a way its model did not predict. The instruments performed; what failed was around them: the cash, the exit, the information, the venue, the buyer's ability to price, or the decision not to hedge at all. Across the cases four mechanisms recur, and the instruments do not.

Cash differs from value. Metallgesellschaft's stack hedge was flat in value at every price and catastrophic in cash, because margin is paid now and the supply profit arrives over years. The UK LDI schemes and the European generators met the same shape in the same quarter of 2022, a hedge margined daily against a hedged item that is not, which suggests the exposure is created by market structure. AIG met it in another form, with collateral called on spread widening and no defaults at all.

Size and leverage change the exit, and someone must see the whole. LTCM, Amaranth, and Archegos failed at the point of liquidation rather than of valuation. Archegos's prime brokers each margined their own slice of a position no one saw whole; at Barings and Société Générale a trader defeated the controls, and the ratio of the hidden position to equity decided which firm survived. Portfolio insurance in 1987 was the market-wide version: a hedge rule shared widely enough becomes the flow and deepens the decline it insures against.

The guarantee is bounded, and measures can bend. When nickel more than tripled in two days the exchange voided executed trades, so a central counterparty's guarantee depends on its resources and its governance. An approved model change halved the London Whale's VaR with no position changed. A fifth mechanism sits apart: Orange County could not price what it bought, though nothing was concealed.

The most recent case reverses the usual story. Silicon Valley Bank failed because it left its rate risk unhedged, so a reading that treats derivatives as the source of danger has the sign wrong. The lesson is to separate the instrument from the failure. In most cases the facts needed to prevent the loss already existed inside the firm, and what failed was escalation. The control failures produced durable reform; the measurement failures produced the least.

FURTHER READING

Most of these events produced an official inquiry or a commissioned review, and those reports are the primary sources. On Barings, the place to start is the Board of Banking Supervision's 1995 inquiry; on 1987, the Brady Commission's report of 1988, which Section 16.2 also draws on. On the London Whale, JPMorgan's own Management Task Force report and the

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US Senate Permanent Subcommittee on Investigations' case history, both of 2013, are worth reading together, since one is the bank's account and the other a legislative investigation of it. On Archegos, the Paul, Weiss report prepared for Credit Suisse (2021) is the primary source, particularly on what the bank's own risk function reported while the position was building. For the 2022 episodes, Oliver Wyman's independent review of the nickel suspension, prepared for the LME (2023), and the Bank of England's *Financial Stability Report* of December 2022 on the LDI intervention both state what the authorities did not know at the time, which is the part worth reading. On Silicon Valley Bank, the Federal Reserve's April 2023 review of its own supervision is the source, and it is explicit about examination failures. On AIG, the Financial Crisis Inquiry Commission's report (2011) sets out the collateral calls and the downgrade. On Metallgesellschaft, Culp and Miller (1995) and Edwards and Canter (1995) are worth reading for the disagreement rather than the conclusion: whether the hedge was sound depends entirely on whether one counts the funding requirement as part of the strategy, and the two sides of that argument are the two halves of Section 40.1.

PRACTICE QUESTIONS

40.1 Using Example 40.1, explain why the stack hedge is flat in value and catastrophic in cash, and identify which leg produces the cash demand.

40.2 Using Figure 40.1, state the peak funding gap and when it occurs, and explain why the strategy can fail between the two points even though both curves end at the same total.

40.3 Explain how a stack-and-roll converts a flat-price hedge into a calendar spread, and state the first-year cost in Example 40.1's contango case.

40.4 Explain what Barings and Société Générale have in common, and why the same failure destroyed one firm and not the other.

40.5 Using Example 40.3, compute the move that exhausts LTCM's equity, and explain why this makes the strategy a view about the path rather than about value.

40.6 Using the Amaranth figures, compute the days required to exit at a 10% participation rate, and explain why the measure's two assumptions, its one-day horizon and prices unmoved by the seller, fail simultaneously.

40.7 Using Example 40.4, explain why each prime broker could judge its own slice adequately margined while the system was not.

40.8 Explain what assumption of Chapter 2 the nickel case tested, and what the risk that materialized actually was.

40.9 Using Example 40.5, explain how a scheme's funding position could improve at every step while it was unable to meet collateral calls.

40.10 Explain why AIG's collateral demands required no defaults at all, and what distinction of Chapter 25 the case turns on.

40.11 Explain what changed and what did not change when the credit index desk adopted a new VaR model, and state the loss as a multiple of each reported figure.

40.12 Using Example 40.6, explain how portfolio leverage and instrument leverage multiply, infer the pool's effective duration from its $1.64 billion loss on a 300 bp rise, and compute the loss on a 200 bp rise at that duration.

40.13 Explain why nothing was concealed in the Orange County case and the buyer still lost, and what this implies about the limits of disclosure.

40.14 Explain why a European generator that had sold power forward faced margin calls while

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its economics improved, and compute the demand on 50 TWh for a move from 50 to 400 EUR/MWh.

40.15 Explain why a valuation that cannot be independently checked responds to pressure, and why the pressure is asymmetric.

40.16 Explain the case study's finding that the failure is rarely detection, and why so little of the resulting reform is quantitative.

40.17 Using Example 40.7, compute the economic loss on a \$120bn portfolio at duration 5.6 for a 300 bp rise, and state it as a share of a \$16bn equity base.

40.18 Explain the role of held-to-maturity accounting in the same case, and why a loss disclosed in the footnotes never reached capital, yet a sale and a capital raise put it in front of depositors.

40.19 Name the four mechanisms that recur across these cases, and give one case for each.

40.20 Explain which category of failure produced durable reform and which produced the least, with an example of each.

40.21 Using Example 40.2 and Figure 40.2, compute the futures the programs must sell as the index falls from 100 to 90, and explain why the selling per index point peaks near the floor and slows once the delta approaches -1.

FURTHER QUESTIONS

40.22 Example 40.1 separates the value and cash profiles of a hedge. Construct a general measure of a hedge's funding requirement (the distribution of cumulative cash demanded before the offsetting gain is realized) and apply it to the LDI position of Example 40.5.

40.23 Example 40.4 shows margin adequate bilaterally and inadequate in aggregate. Design a reporting mechanism that would have revealed the total, state what each party would have to disclose and to whom, and determine whether the disclosure would itself have moved the market.

40.24 Section 40.5 treats exit time as a function of position share. Build a liquidation-cost model with price impact proportional to participation, determine the optimal unwind schedule for a position at 40% of open interest, and compare the resulting cost with the position's reported VaR.

40.25 The nickel case ended in trade cancellation. Design a CCP recovery arrangement that avoids voiding executed trades, state who funds it, and determine whether any participant would rationally join a clearing house on those terms.

40.26 Section 40.14 argues the accounting treatment concealed the exposure. Model a bank's optimal hedging decision under held-to-maturity accounting versus full fair value, and determine the conditions under which the accounting choice alone flips the hedging decision.

40.27 Section 40.16 argues that measurement failures produced the least reform. Take the concentration gap specifically, propose a supervisory measure that would capture it, and determine what data would be required and from whom.

40.28 Every case in this chapter was reconstructed after the fact. Choose one and identify the earliest point at which the information available inside the firm would have been sufficient to act, then state what report would have had to exist for someone to have seen it.

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